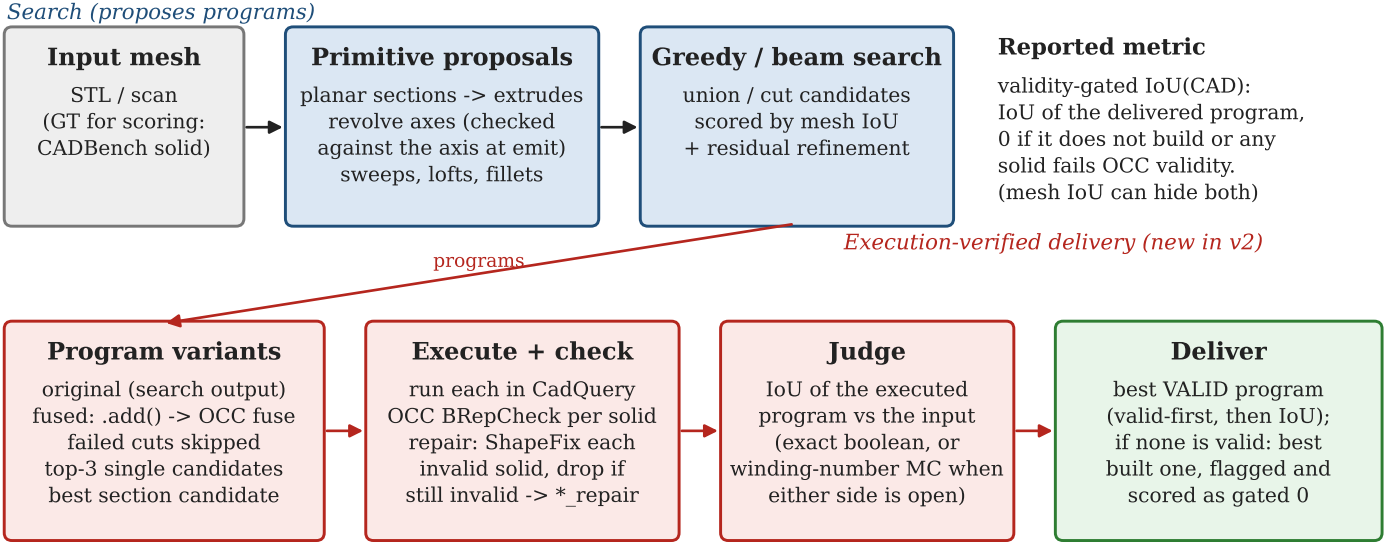
Search (proposes programs)
Input mesh
STL / scan
(GT for scoring:
CADBench solid)
Primitive proposals
planar sections -> extrudes
revolve axes (checked
against the axis at emit)
sweeps, lofts, fillets
Greedy / beam search
union / cut candidates
scored by mesh IoU
+ residual refinement
Reported metric
validity-gated IoU(CAD):
IoU of the delivered program,
0 if it does not build or any
solid fails OCC validity.
(mesh IoU can hide both)
Execution-verified delivery (new in v2)
programs
Program variants
original (search output)
fused: .add() -> OCC fuse
failed cuts skipped
top-3 single candidates
best section candidate
Execute + check
run each in CadQuery
OCC BRepCheck per solid
repair: ShapeFix each
invalid solid, drop if
still invalid -> *_repair
Judge
IoU of the executed
program vs the input
(exact boolean, or
winding-number MC when
either side is open)
Deliver
best VALID program
(valid-first, then IoU);
if none is valid: best
built one, flagged and
scored as gated 0